

Linear Algebra 101 — Part 5: Determinants

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In this story, we are going to cover possibly one of the most important concepts in Linear Algebra, “**Determinants**”!

Materials covered in this article

- Determinants and it's general idea
- Properties of Determinants
- Derivation of 2 x 2 determinants using the properties

Introduction to determinants

Determinants is a unique concept that memorizing the formula is rather simple, but understanding its meaning and true potential is often more challenging. In short, “determinant” is the scale factor for the area or volume represented by the column vectors in a square matrix. To understand this, I can't stress enough how important it is to watch this amazing video from 3Blue1Brown.

Make sure to just spend 10 mins watching this video to get general idea on what determinant is.

Determinant - Wikipedia

Determinants occur throughout mathematics. For example, a matrix is often used to represent the coefficients in a...

en.wikipedia.org

Now you have a general idea about what the determinant is. Let's take a look at the formula now.

Determinants

$$\det A = |A| = \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$$

($\times \mathbb{R}^{2 \times 2}$ case)

This is the two by two case and the general formula is rather complexing so I'll just leave it to the wiki page for you to check

Determinant - Wikipedia

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Determinants and its properties

This is actually the main topic we want to discuss today. The properties of the determinant.

You can see the complete lecture from Dr.Strang from MIT on this at Lecture 18.

I'm basically summarizing and adding some more information so that it's easier to understand.

Properties of Determinants

$$1. \det I = 1$$

Example :

$$\det \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} = 1 \cdot 1 - 0 \cdot 0 = 1$$

If you watched the 1st video posted on this story, you can understand this first property. If it's the identity matrix that we are talking about, the column vectors are orthogonal to each other and there's nothing to scale. That's why the determinant is always 1.

$$2. \text{ Exchange rows} \\ \hookrightarrow \text{Reverse sign of the det.}$$

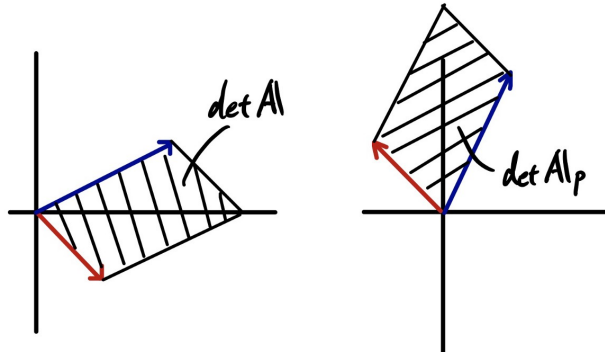
$$\det(\text{Permutation Matrix}) \\ = \begin{cases} 1 & \text{if \# of exchange is even.} \\ -1 & \text{if \# of exchange is odd.} \end{cases}$$

The 2nd property is that when you exchange any rows, it will flip the sign of the determinant.

If you remember the the video from 3Blue1Brown, you could understand the following example. Basically, by exchanging rows, the arrows flip its positions, thus resulting in changing it's signs.

$$A = \begin{bmatrix} 1 & 2 \\ -1 & 1 \end{bmatrix}$$

$$A_p = \begin{bmatrix} -1 & 1 \\ 1 & 2 \end{bmatrix}$$



3a. If a row is multiplied by a constant,
You can take it outside the det.

$$\begin{vmatrix} ta & tb \\ c & d \end{vmatrix} = t \begin{vmatrix} a & b \\ c & d \end{vmatrix}$$

Let's take an easy example here again.

Example

$$\det A = \begin{vmatrix} 2 & 4 \\ 1 & 3 \end{vmatrix} = 2 \cdot 3 - 4 \cdot 1 \\ = 6 - 4 = 2$$

$$\begin{vmatrix} 2 & 4 \\ 1 & 3 \end{vmatrix} = 2 \begin{vmatrix} 1 & 2 \\ 1 & 3 \end{vmatrix} \\ = 2(1 \cdot 3 - 2 \cdot 1) \\ = 2(3 - 2) = 2 \cdot 1 = 2$$

Similar to the above, there's another property in 3.

3b. If something is added to the row,
You can separate the det.

$$\begin{vmatrix} a+a' & b+b' \\ c & d \end{vmatrix} = \begin{vmatrix} a & b \\ c & d \end{vmatrix} + \begin{vmatrix} a' & b' \\ c & d \end{vmatrix}$$

Let's take another easy example for this property too.

Example

$$\det A = \begin{vmatrix} 1 & 3 \\ 5 & 8 \end{vmatrix} = 1 \cdot 8 - 3 \cdot 5 = 8 - 15 = -7$$

$$\begin{aligned} \begin{vmatrix} 1 & 3 \\ 5 & 8 \end{vmatrix} &= \begin{vmatrix} 1 & 3 \\ 5 & 0 \end{vmatrix} + \begin{vmatrix} 1 & 3 \\ 0 & 8 \end{vmatrix} \\ &= (0 - 15) + (8 - 0) \\ &= -15 + 8 = -7 \end{aligned}$$

The 4th property can be derived from the 2nd property we already described.

4. Two equal rows $\rightarrow \det = 0$

$\hookrightarrow$ From the property 2

$\hookrightarrow$ exchange the two equal rows

$\hookrightarrow$ Still the same original matrix

$\hookrightarrow$ Conflict in the change of the signs.

$\hookrightarrow$ det must be zero.

From here, it's easier to show why using the properties that we already covered. The 5th property can be shown using multiple properties.

5. Subtract $l \times \text{row}_i$ from row_k
 $\hookrightarrow \det$ does not change.

$$\begin{aligned} & \begin{vmatrix} a & b \\ c - la & d - lb \end{vmatrix} \\ &= \begin{vmatrix} a & b \\ c & d \end{vmatrix} + \begin{vmatrix} a & b \\ -la & -lb \end{vmatrix} \quad \left. \begin{array}{l} \text{Property 3b} \\ \text{Property 3a} \end{array} \right\} \\ &= \begin{vmatrix} a & b \\ c & d \end{vmatrix} - l \begin{vmatrix} a & b \\ a & b \end{vmatrix} \\ & \quad \quad \quad \hookrightarrow \text{Property 4} \rightarrow 0 \\ &= \begin{vmatrix} a & b \\ c & d \end{vmatrix} \end{aligned}$$

Here's some concrete example.

Concrete Example of 5.

$$A = \begin{bmatrix} 1 & 2 \\ 5 & 9 \end{bmatrix} \quad \det A = 1 \cdot 9 - 2 \cdot 5 = 9 - 10 = -1$$

$$A' = \begin{bmatrix} 1 & 2 \\ 5 - 4 \cdot 1 & 9 - 4 \cdot 2 \end{bmatrix} \quad (l = 4)$$

$$\det A' = \begin{vmatrix} 1 & 2 \\ 1 & 1 \end{vmatrix} = 1 \cdot 1 - 2 \cdot 1 = -1 = \det A$$

6. Row of zeros $\rightarrow \det = 0$

$\hookrightarrow$ Related to Property 4.

Using Property 5., 4. situation becomes 6.

$$\begin{vmatrix} 1 & 2 \\ 1 & 2 \end{vmatrix} \xrightarrow{\text{Property 4}} \begin{vmatrix} 1 & 2 \\ 0 & 0 \end{vmatrix} \xrightarrow{\text{Property 5}} \begin{vmatrix} 1 & 2 \\ 0 & 0 \end{vmatrix} \xrightarrow{\text{Property 6}} 0$$

7. If A is a triangular matrix, the $\det$ is the product of the diagonal (pivots) entries.

$$\det A = \begin{vmatrix} d_1 & - & - & - & - \\ & d_2 & - & - & - \\ & & d_3 & - & - \\ & & & \ddots & \\ 0 & & & & d_n \end{vmatrix} = d_1 d_2 d_3 \cdots d_n = \sum_{i=1}^n d_i$$

Why?

$$\hookrightarrow \text{rref} \begin{vmatrix} d_1 & 0 & 0 & \cdots & 0 \\ 0 & d_2 & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & & d_n \end{vmatrix} = d_1 d_2 \cdots d_n \begin{vmatrix} 1 & 0 & 0 & \cdots & 0 \\ 0 & 1 & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \end{vmatrix}$$

Property 3a

$$= d_1 d_2 \cdots d_n \det I$$

Property 1 $\hookrightarrow = d_1 d_2 \cdots d_n$

The last three properties are pretty important as it comes back again and again in the later stories.

For example, property 8 is so important because it's used to check if the matrix is invertible or not.



$$\begin{array}{ll} 8. & \det A = 0 \quad \text{when } A \text{ is singular.} \\ & \det A \neq 0 \quad \text{when } A \text{ is invertible.} \\ & \hspace{15em} (\text{regular}) \end{array}$$

Property 9 comes in handy when it is difficult to calculate some determinants.



$$9. \det(A|B) = (\det A)(\det B)$$

Example: $\det A^{-1} \rightarrow \text{Use } AA^{-1} = I$

$$\det(AA^{-1}) = \det I = 1$$

$\downarrow$ $\nearrow$

Property 9 Property 1

$$(\det A)(\det A^{-1}) = 1$$

$$\therefore \det A^{-1} = \frac{1}{\det A}$$

Property 10 is good to remember too.



$$10. \det A^T = \det A$$

So that's it for the properties of the determinants. Let's use these to derive the determinants equation of 2 x 2 case.

Deriving the original formula for determinants 2 x 2 case

First, let's recap what the formula was.

$$\begin{array}{l} \text{Derivation of Determinant Formula} \\ A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \quad \det A = \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc \end{array}$$

To get this formula, we are going to use quite a lot of the properties that we already discussed.

$$\begin{aligned}
 \begin{vmatrix} a & b \\ c & d \end{vmatrix} &= \begin{vmatrix} a & 0 \\ c & d \end{vmatrix} + \begin{vmatrix} 0 & b \\ c & d \end{vmatrix} \quad (3b) \\
 &= \begin{vmatrix} a & 0 \\ c & 0 \end{vmatrix} + \begin{vmatrix} a & 0 \\ 0 & d \end{vmatrix} + \begin{vmatrix} 0 & b \\ 0 & d \end{vmatrix} + \begin{vmatrix} 0 & b \\ c & 0 \end{vmatrix} \quad (3b) \\
 \text{Here } \begin{vmatrix} a & 0 \\ c & 0 \end{vmatrix} \text{ \& } \begin{vmatrix} 0 & b \\ 0 & d \end{vmatrix} \text{ is Singular.} \quad (*) \\
 \text{Proof } \begin{vmatrix} a & 0 \\ c & 0 \end{vmatrix} &= \begin{vmatrix} a & 0 \\ 0 & 0 \end{vmatrix} = 0 \quad (6) \\
 &\quad \xrightarrow{\text{ref}} \text{Contains rows with zeros}
 \end{aligned}$$

Then restarting from the equation that we just started...

$$\begin{aligned}
 (*) &= \begin{vmatrix} a & 0 \\ 0 & d \end{vmatrix} + \begin{vmatrix} 0 & b \\ c & 0 \end{vmatrix} \\
 &= \begin{vmatrix} a & 0 \\ 0 & d \end{vmatrix} - \begin{vmatrix} c & 0 \\ 0 & b \end{vmatrix} \quad (2) \text{ Exchange rows} \\
 &= ad \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} - bc \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} \quad (3a) \\
 &= ad \det I - bc \det I \\
 &= ad - bc \quad (1)
 \end{aligned}$$

Great! Now I hope you know a lot about the properties and the determinant itself!

Update Feb 27th 2019

I found a great web resource where you can interactively learn linear algebra. Check this web page for learning determinants interactively:

Chapter 7: Determinants (Immersive Linear Algebra)

`\begin{equation} \backslashbegin{array}{llr} (i) \& \det( \backslashmx{I} ) = 1 \& \\ \backslashspc\text{(determinant of unit matrix)} \& \backslash\ (ii) \& \dots \end{array}`

immersivemath.com

Summary

Determinant is an important scale in linear algebra. That's why it has a lot of properties. You don't need to remember everything line by line. First, try to get the ideas. Then play with the properties. With examples you created. Then it should just stick to your mind so that you could use it yourself. Just like learning how to use some tools like a saw.

I hope this helps! See you next time~ (· ω ·)

